Econometrics I
PPS-02
Multiple Choice Questions
(chapters 3-4 (F test excluded))
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1. Fill in the blank. **The Gauss-Markov theorem will not hold if ..... .**
   A. the error term has the same variance given any values of the explanatory variables
   B. the error term has an expected value of zero given any values of the independent variables
   C. the regression model relies on the method of random sampling for collection of data
   D. the independent variables have exact linear relationship among them
   E. the model is linear in parameters.

2. Fill in the blanks. **The Gauss-Markov theorem states that OLS estimator is the best ..... and ..... estimator among the set of all linear and unbiased estimators. This is sometimes stated as ..... .**
   A. linear / minimum variance / BLUME
   B. minimum variance / unbiased / MLUE
   C. linear / unbiased / LUBE
   D. nonlinear / unbiased / NLUE
   E. linear / unbiased / BLUE

3. Fill in the blanks. **When an important variable is omitted, OLS estimator becomes ..... This is known as .... bias .**
   A. biased / omitted variable
   B. biased / irrelevant variable
   C. biased / measurement error
   D. biased / heteroskedasticity
   E. biased / multicollinearity

4. Fill in the blanks. **When an unimportant (<u>irrelevant</u>) variable is included, OLS estimator is ..... but ..... increases .**
   A. unbiased / bias
   B. biased / bias
   C. biased /variance
   D. unbiased / variance
   E. unbiased / heteroskedasticity

5. Consider the following regression model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + u$$

   If the sample size is $n = 250$, then the degrees of freedom is dof = ...
   A. 242
   B. 243
   C. 249
   D. 250
   E. 257

6. Consider the results of the following regression model:

$$\widehat{\mathrm{y}} = \underset{(10)}{15} + \underset{(0.75)}{1.5}\,\mathrm{x}_1 - \underset{(2.0)}{3.0}\,\mathrm{x}_2 + \underset{(0.28)}{0.45}\,\mathrm{x}_3$$
$$n = 140 \quad R^2 = 0.59$$

You want to test the null hypothesis $H_0 : \beta_{x_2} = -2.0$

The calculated test statistic is ..... and we ..... the null hypothesis at 5% significance level against the two-sided alternative.(Note: For two-sided t-test critical value at 5% is 1.96)

Fill in the blanks.

A. -0.5, reject

B. 2.5, fail to reject

C. -2.5, fail to reject

D. -0.5, fail to reject

E. -2.5, reject

7. Fill in the blanks. **We can rely on ..... Theorem to assume that the error term follows the ..... distribution.**

A. Central Limit / t

B. Gauss-Markov / Nonnormal

C. Gauss-Markov / Normal

D. Central Limit / Nonnormal

E. Central Limit / Normal

8. The following is the R regression output for the PRF:

$$educ = \beta_0 + \beta_1 fatheduc + \beta_2 motheduc + \beta_3 sibs + \beta_4 abil + u$$

Variable definitions: `educ` is the education level measured in years, `fatheduc` and `motheduc` are the education levels of father's and mother's, respectively, both measured in years, `sibs` is the number of siblings, and `abil` is a measure of cognitive ability.

```
lm(formula = educ ~ fatheduc + motheduc + sibs + abil)
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.97212    0.31766  28.244  < 2e-16
fatheduc     0.150      0.030    ...... 2.04e-08
motheduc     0.173      0.029     5.966 1.86e-09
sibs        -0.109      0.028    -3.893 0.000102
abil         0.500      0.025    20.000  < 2e-16
---

Residual standard error: 1.773 on 1220 degrees of freedom
Multiple R-squared:  0.50
```

What is the t statistic for the fatheduc? Is it statistically significant?

A. $t = 5$. fatheduc is statistically insignificant at 5% level.

B. $t = 5$. fatheduc is statistically significant at 1% level.

C. $t = 5$. We don't know if fatheduc is significant because the critical value is not given.

D. $t = 5$. fatheduc is statistically insignificant at 1% level.

E. $t = 4.5$. fatheduc is statistically significant at 1% level.

9. Consider the following regression model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$

Suppose that we are interested in testing $H_0 : \ \beta_1 = 0$ against the two-sided alternative $H_1 : \ \beta_1 \neq 0$. We computed the t-statistic and its p-value is found to be 0.06. **Which of the following is the true interpretation of the p-value?**

A. We reject the null hypothesis at 5% level but not at 10% level.

B. We reject the null hypothesis at 1% level.

C. The smallest significance level that leads to the rejection of H_0 is 0.06, i.e., 6%. This means that we do not reject the null at 10%.

D. The smallest significance level that leads to the rejection of H_0 is 0.06, i.e., 6%. This means that we reject the null at 5%.

E. The smallest significance level that leads to the rejection of H_0 is 0.06, i.e., 6%. This means that we do not reject the null at 5%.

10. Suppose that the 95% confidence interval for a parameter is found to be $[-0.88, \ 0.98]$. **What can we say about the statistical significance of that parameter?**

A. Since the interval includes 0, the effect is significant.

B. Since the interval includes 0.90, the effect is significant.

C. Since the interval includes 0, the effect is insignificant.

D. Since the interval does not include 0, the effect is insignificant.

E. Since the interval does not include 0, the effect is significant.

11. Suppose that the 95% confidence interval for a parameter is found to be $[-0.88, \ 0.98]$. The 5% two-sided critical value is 1.96. **Can you find the OLS estimate $\hat{\beta}$ based on the given information?**

A. No, we need to know the standard error.

B. No, we need to know the full OLS results.

C. Yes, $\hat{\beta} = 0$

D. Yes, $\hat{\beta} = 0.05$

E. Yes, $\hat{\beta} = 0.5$